

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 1**

**COURSE NAME: COMPUTER NETWORKS**

**COURSE CODE: CSA0704**

### **COURSE OUTCOME COVERED**

**CO5:** Measure network performance using key metrics with simulation tools

**(BL4, BL5)**

***Assessment Tool Weightage: 40%***

**QUESTIONS: 3**

**Total Marks:30**

### **Annexure A**

#### **Constraint-Based Problem Solving**

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and

authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

| <b>RUBRICS<br/>FOR<br/>CALCULATION<br/>Criteria</b> | <b>Weightage<br/>(%)</b> | <b>Exemplary<br/>(4)</b>                                                          | <b>Proficient<br/>(3)</b>                                    | <b>Developing<br/>(2)</b>                             | <b>Beginning<br/>(1)</b>                                   |
|-----------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------|
| Understanding of Problem & Constraints              | 20%                      | Clearly identifies all constraints and fully understands the problem requirements | Identifies most constraints with minor gaps in understanding | Identifies some constraints but misses key aspects    | Shows little or no understanding of constraints or problem |
| Problem Decomposition                               | 15%                      | Breaks problem into logical, well-structured sub-problems effectively             | Breaks problem into sub-parts with some logical flow         | Attempts decomposition but lacks clarity or structure | No clear decomposition of the problem                      |

|                                        |     |                                                                           |                                                      |                                                     |                                                |
|----------------------------------------|-----|---------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|------------------------------------------------|
| Application of Constraints in Solution | 20% | Applies all constraints correctly and consistently in solution design     | Applies most constraints correctly with minor errors | Applies some constraints but with noticeable errors | Fails to apply constraints appropriately       |
| Logical Reasoning & Strategy           | 15% | Uses strong, clear, and efficient reasoning strategies                    | Uses generally sound reasoning with minor gaps       | Reasoning is partially correct but inconsistent     | Reasoning is unclear or incorrect              |
| Solution Accuracy & Feasibility        | 15% | Solution is fully correct, feasible, and optimized                        | Solution is mostly correct and feasible              | Solution has errors or is partially feasible        | Solution is incorrect or not feasible          |
| Creativity & Innovation                | 5%  | Demonstrates highly innovative and unique approach                        | Shows some creativity in approach                    | Limited creativity shown                            | No creativity; relies on basic or copied ideas |
| Clarity of Presentation                | 10% | Solution is clearly presented with excellent organization and explanation | Presentation is clear with minor issues              | Presentation lacks clarity or organization          | Poorly presented and difficult to understand   |



## Question 1: Remote Branch Office Connectivity

A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. Recommend a suitable internet connection, networking devices, and security mechanisms, justifying the recommendations against bandwidth, reliability, cost, and future scalability.

### 1. Understanding of the Problem & Constraints

The central constraint is the absence of wired broadband infrastructure in a remote rural area, which rules out fiber/leased-line options that would normally be the default for a corporate branch. Layered on top of this is a business requirement for near-zero downtime (cloud apps and video conferencing are both latency- and availability-sensitive), a limited connectivity budget, and an implicit future-scalability requirement as the branch grows.

These constraints pull in different directions: high reliability and bandwidth typically mean higher cost, so the design must find the most cost-efficient way to meet an uninterrupted-operations requirement rather than simply buying the most expensive available link.

### 2. Problem Decomposition

- Sub-problem 1 - Access technology selection: identify which last-mile technology is even available in a remote area (VSAT satellite, 4G/5G cellular broadband) and compare them on bandwidth and cost.
- Sub-problem 2 - Redundancy/reliability design: since no single rural link is fully reliable, determine how to combine two access technologies for automatic failover.
- Sub-problem 3 - On-premise networking equipment: select the router/gateway class needed to manage multiple WAN links intelligently.
- Sub-problem 4 - Security and HQ connectivity: determine how to secure branch traffic and establish confidential communication with headquarters over a public/shared medium.
- Sub-problem 5 - Scalability: ensure the chosen architecture can add bandwidth or a second office without a redesign.

### 3. Constraint Analysis

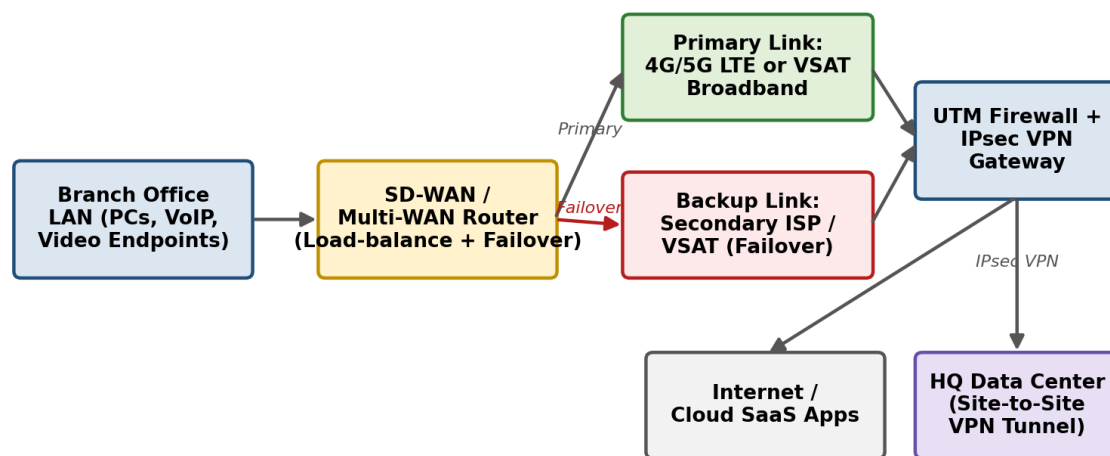
| Constraint                        | Impact on Design                               | How It Is Addressed                                                                                             |
|-----------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| No wired broadband available      | Eliminates fiber/leased-line as primary option | Use 4G/5G LTE (or VSAT where cellular is unavailable) as the primary WAN link                                   |
| Uninterrupted operations required | Single link = single point of failure          | Dual-WAN SD-WAN router with automatic failover to a second ISP/VSAT link                                        |
| Limited budget                    | Cannot afford premium leased-line SLA          | Consumer/business-grade LTE plans + a lower-cost secondary VSAT/backup link, avoiding costly dedicated circuits |
| Cloud apps & video conferencing   | Needs adequate throughput and low jitter       | SD-WAN traffic shaping/QoS to prioritize video and SaaS traffic over bulk downloads                             |
| Secure HQ communication           | Data crosses a public/cellular network         | Site-to-site IPsec VPN tunnel + UTM firewall at the branch                                                      |
| Future scalability                | Branch may grow or add sites                   | SD-WAN architecture allows adding WAN links or new branch sites via centralized policy, without re-architecting |

## 4. Proposed Solution & Network Diagram

The recommended design uses a dual-WAN SD-WAN router as the branch gateway, terminating a primary 4G/5G LTE (or VSAT, where cellular coverage is insufficient) connection and a secondary, independent ISP/VSAT link purely for failover. The SD-WAN router continuously monitors both links' latency/packet loss and automatically shifts traffic to the healthy link within seconds of a failure, directly satisfying the 'uninterrupted operations' requirement without paying for an expensive dedicated leased line.

A UTM (Unified Threat Management) firewall sits behind the router to provide stateful inspection, intrusion prevention, and the endpoint for a site-to-site IPsec VPN tunnel to HQ, so all branch-to-HQ traffic is encrypted end-to-end even though it traverses commodity internet links. Cloud/SaaS application traffic and video-conferencing traffic are prioritized via SD-WAN QoS policies so that a bandwidth-heavy background task cannot degrade a live video call.

### Recommended Network Design - Remote Branch Office



Key: Green = primary path, Red = backup/failover path, Yellow = SD-WAN router (auto path selection)

Figure 1: Recommended dual-WAN SD-WAN architecture with primary/backup links and IPsec VPN to HQ.

## 5. Logical Reasoning & Strategy

The reasoning strategy follows a constraint-satisfaction approach: first eliminate infeasible options (wired broadband), then select the cheapest technology that meets the bandwidth floor for cloud apps and video (cellular LTE, which now commonly offers tens of Mbps - sufficient for HD video conferencing), then add the minimum redundancy needed to satisfy the availability requirement (a second, independent link rather than a fully duplicated expensive circuit), and finally layer security on top without altering the underlying access technology choice.

This ordering matters: solving reliability before cost (e.g., defaulting to two premium VSAT links) would blow the budget constraint, while solving cost before reliability would violate the uninterrupted-operations requirement - the SD-WAN dual-link approach is the point in the trade-off space that satisfies both simultaneously.

## 6. Solution Accuracy & Feasibility

The solution is feasible with commercially available, well-established technology: 4G/5G LTE routers with dual-SIM/dual-WAN failover, VSAT as a rural fallback, and SD-WAN/UTM appliances are standard products from vendors such as Cisco Meraki, Fortinet, and Peplink, meaning the design can be deployed with off-the-shelf hardware rather than custom engineering. Bandwidth is adequate for the stated workload (cloud apps + video conferencing typically need well under 50 Mbps), reliability is addressed via automatic failover, and total cost stays well below a dedicated leased-line alternative.

## 7. Creativity & Innovation

The innovative element is treating connectivity as a software-defined, policy-driven resource (SD-WAN) rather than a fixed physical circuit - this lets the branch's network administrator add, remove, or re-prioritize WAN links purely through configuration as the business scales, and allows the same architecture to be replicated instantly at future remote branches, turning a one-off connectivity problem into a reusable, centrally managed pattern.

## Question 2: Campus-Wide Wireless Network Design

A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels, supporting a large number of simultaneous users while minimizing interference and ensuring secure access, under a constraint of a limited number of deployable access points. Propose an efficient wireless design including AP placement, frequency band selection, and authentication methods, explaining the coverage/performance/security/cost balance.

### 1. Understanding of the Problem & Constraints

The problem has a hard capacity constraint (limited number of APs) set against a large, unevenly distributed user population (3,000+ students across four distinct area types with very different density and usage patterns), plus co-existing requirements for minimized RF interference and secure authenticated access - all without exceeding budget.

The key insight is that not all areas have the same requirement: classrooms and labs need high client density support in a small footprint, while hostels need broad coverage over a large area with comparatively fewer simultaneous active users per AP - treating all zones identically would either overspend on low-density areas or under-provision high-density ones.

### 2. Problem Decomposition

- Sub-problem 1 - Zone classification: classify each area type (classrooms, labs, library, hostels) by expected simultaneous client density and required coverage footprint.
- Sub-problem 2 - AP placement & count allocation: distribute the limited AP budget preferentially toward high-density zones while ensuring hostels still get baseline coverage.
- Sub-problem 3 - Frequency band and channel planning: choose 2.4 GHz vs 5 GHz per zone and plan non-overlapping channels to minimize co-channel interference.
- Sub-problem 4 - Centralized management: introduce a Wireless LAN Controller (WLC) to coordinate RF power/channel assignment automatically across all APs.
- Sub-problem 5 - Authentication and security: select an enterprise-grade authentication method that scales to thousands of students without per-device manual configuration.

### 3. Constraint Analysis

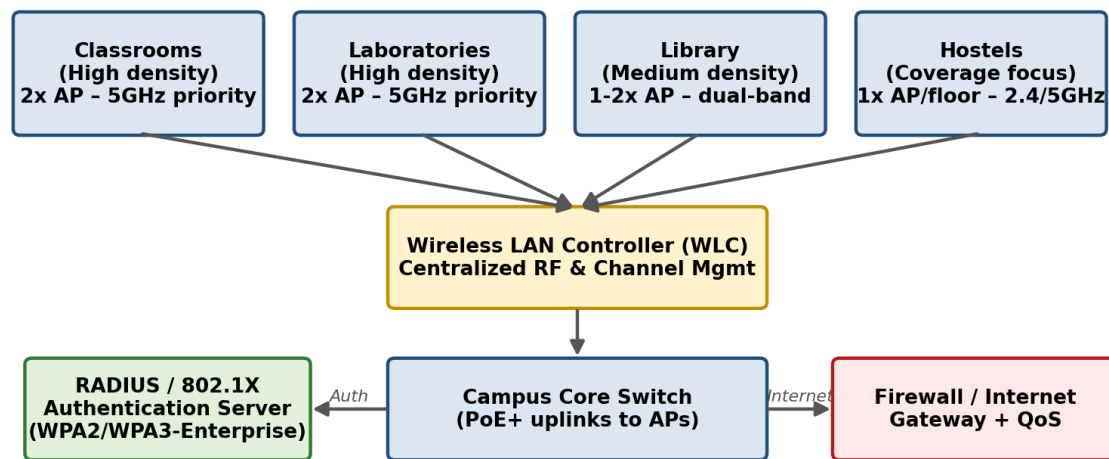
| Constraint                              | Impact on Design                                      | How It Is Addressed                                                                                           |
|-----------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Limited number of APs                   | Cannot blanket every square meter uniformly           | Density-weighted allocation: more APs in classrooms/labs, fewer, higher-power APs for hostel corridors/floors |
| 3,000+ simultaneous users               | Risk of AP/channel congestion                         | 5 GHz prioritized in high-density zones (more non-overlapping channels, less legacy-device congestion)        |
| Interference minimization               | Overlapping APs on same channel degrade throughput    | Centralized WLC performs automatic channel/power planning (RRM) across all APs                                |
| Secure access for thousands of students | Manual per-user configuration is unscalable           | WPA2/WPA3-Enterprise with 802.1X + RADIUS, using existing university credentials (single sign-on)             |
| Budget constraint                       | Cannot deploy unlimited APs or a separate auth system | Reuse of existing directory/credential system for RADIUS; AP count optimized rather than maximized            |

## 4. Proposed Solution & Network Diagram

The design allocates access points by expected client density rather than evenly by area: classrooms and laboratories receive a higher AP-to-room ratio operating predominantly on the 5 GHz band (which offers many more non-overlapping channels - up to 24 in most regulatory domains versus effectively 3 usable channels at 2.4 GHz - making it far better suited to dense, simultaneous-client environments). The library, with moderate and steadier occupancy, uses a dual-band configuration, while hostels use fewer APs per floor operating on both bands, since 2.4 GHz's longer range is better suited to covering corridors and rooms with fewer walls-per-AP.

All APs are centrally managed by a Wireless LAN Controller, which runs automatic Radio Resource Management (RRM) to assign non-overlapping channels and balance transmit power across adjacent APs, directly addressing the interference-minimization requirement without manual per-AP tuning. Authentication is handled via WPA2/WPA3-Enterprise with 802.1X, backed by a RADIUS server integrated with the university's existing student directory (LDAP/Active Directory), so every student uses their existing university login rather than a shared or manually distributed key.

### Campus-Wide Wireless Network Design



*Design balances cost (fewer, well-placed APs), performance (5GHz for dense areas, channel planning to avoid interference), security (WPA2/WPA3-Enterprise via RADIUS), and coverage (2.4GHz for hostel/long-range areas).*

Figure 2: Campus wireless design showing zone-based AP allocation, centralized WLC, RADIUS authentication, and core connectivity.

## 5. Logical Reasoning & Strategy

The reasoning strategy prioritizes solving the hardest constraint first: with a fixed AP budget, the highest-value allocation is achieved by matching AP density to actual usage density (classrooms/labs) rather than to raw floor area, since under-provisioning a lecture hall causes many more simultaneous connection failures than under-provisioning a lightly used corridor.

Frequency-band selection follows directly from this density analysis - high-density zones need 5 GHz's greater channel capacity, while low-density, larger-footprint zones benefit more from 2.4 GHz's superior range - and centralizing RF management through a WLC turns what would otherwise be a manual, error-prone channel-planning exercise into an automated, self-optimizing system, which also keeps ongoing operational cost low.

## 6. Solution Accuracy & Feasibility

The solution is feasible using standard enterprise Wi-Fi products (e.g., Cisco/Aruba/Ubiquiti WLC + AP ecosystems support exactly this density-based deployment model), and 802.1X/RADIUS integration with existing university credentials is a well-established pattern (eduroam uses this exact mechanism across thousands of institutions), so no unproven technology is



required. The AP count constraint is respected by allocation rather than violated, and the design scales - additional APs can be added to any zone later without redesigning the authentication or channel-planning approach.

### 7. Creativity & Innovation

The innovative aspect is applying density-aware, zone-differentiated AP allocation and band selection instead of a one-size-fits-all deployment - this squeezes more effective capacity out of a fixed AP budget than uniform placement would, and adopting an eduroam-style federated RADIUS authentication approach means the university could in future extend secure roaming access to visiting researchers from partner institutions with no additional infrastructure.

## Question 3: Secure Segmented Network for a Financial Institution

A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services, without significantly affecting network performance or increasing operational costs. Design a secure network architecture selecting appropriate segmentation, firewall deployment, authentication methods, and intrusion detection, justified against security, performance, compliance, and budget constraints.

### 1. Understanding of the Problem & Constraints

This problem has competing constraints that are unusually tightly coupled: strong security controls (segmentation, firewalls, IDS, strict authentication) typically add processing overhead and administrative cost, yet the requirement explicitly caps both performance impact and operational cost, while regulatory compliance (a near-mandatory constraint in financial services, e.g., PCI-DSS-style requirements) sets a non-negotiable security floor that cannot be traded away for cost savings.

The practical resolution is not to weaken security to save cost/performance, but to apply security selectively and architecturally - protecting the highest-value assets (customer data) most strongly while keeping lower-risk traffic (general employee LAN use) on a lighter-weight path, rather than applying maximal security uniformly everywhere.

### 2. Problem Decomposition

- Sub-problem 1 - Asset classification: identify which systems hold the highest-sensitivity data (customer/financial records) versus general-purpose employee systems.
- Sub-problem 2 - Network segmentation design: partition the network into zones (DMZ, employee LAN, restricted customer-data segment, admin/IT) using VLANs so a breach in one zone cannot directly reach another.
- Sub-problem 3 - Firewall placement: decide where perimeter versus internal firewalls are needed to inspect traffic between zones without adding a bottleneck to every internal transaction.
- Sub-problem 4 - Authentication strategy: select an authentication mechanism strong enough for compliance (MFA) without imposing excessive friction on daily employee workflows.
- Sub-problem 5 - Monitoring and compliance: introduce IDS/IPS and centralized logging (SIEM) to detect intrusions and produce the audit trail regulators require.

### 3. Constraint Analysis

| Constraint                        | Impact on Design                                | How It Is Addressed                                                                                                                                                          |
|-----------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Strict security & compliance      | Cannot be compromised for cost/performance      | Non-negotiable baseline: perimeter firewall + IPS, internal firewall + IDS, encrypted restricted VLAN for customer data                                                      |
| No significant performance impact | Deep inspection everywhere adds latency         | Segmentation limits deep inspection to zone boundaries only, not every internal packet; high-value traffic (VLAN 20) gets dedicated inspection, general LAN traffic does not |
| Employee access to sensitive data | Data must remain accessible for legitimate work | Role-Based Access Control (RBAC) + MFA grants access only to authorized roles, not the whole network                                                                         |

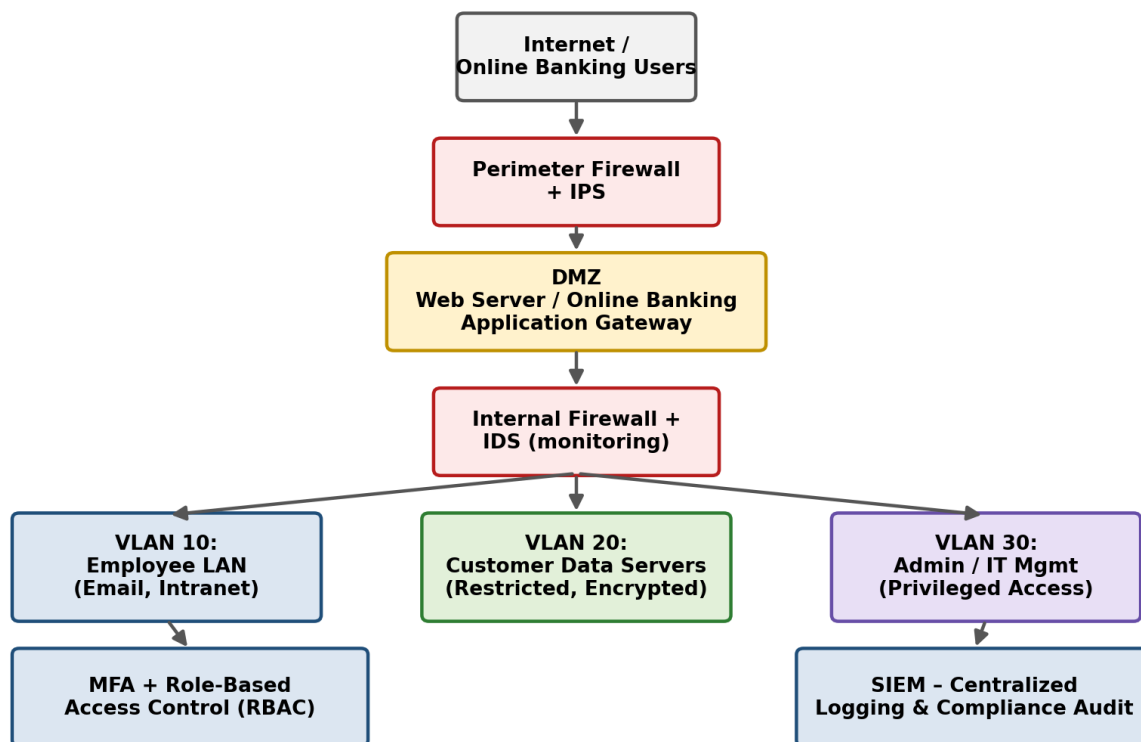
| Constraint                  | Impact on Design                      | How It Is Addressed                                                                                                    |
|-----------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Budget constraint           | Cannot deploy unlimited appliances    | Reuse of a single internal firewall/IDS pair at the critical VLAN-20 boundary instead of dedicated appliances per VLAN |
| Regulatory compliance/audit | Must prove controls and maintain logs | Centralized SIEM aggregates logs from all zones for continuous compliance reporting                                    |

## 4. Proposed Solution & Network Diagram

The architecture places a perimeter firewall with intrusion prevention (IPS) at the internet boundary, in front of a DMZ hosting only the public-facing online-banking web/application gateway - so external users never directly reach the internal network. Behind the DMZ, an internal firewall combined with an IDS inspects traffic crossing into three segmented VLANs: Employee LAN (VLAN 10, general productivity systems), Customer Data Servers (VLAN 20, encrypted and most tightly restricted), and Admin/IT Management (VLAN 30, privileged access only).

Employees authenticate with multi-factor authentication (MFA) combined with role-based access control, so a given employee's credentials only grant access to the VLAN(s) their role requires - a call-center employee, for instance, cannot reach admin/IT management systems even if their account were compromised. All zones feed a centralized SIEM for continuous logging, intrusion alerting, and the audit trail needed for regulatory compliance.

### Secure Segmented Network Architecture - Financial Institution



*Segmentation isolates customer data (VLAN20) behind an internal firewall/IDS layer; MFA and RBAC gate employee access; SIEM provides continuous compliance logging — balancing security, compliance, performance and cost.*

Figure 3: Segmented network architecture with DMZ, internal firewall/IDS boundary, VLAN isolation, MFA/RBAC, and SIEM logging.

## 5. Logical Reasoning & Strategy

The reasoning strategy applies defense-in-depth combined with proportional control: rather than deploying maximum-strength inspection uniformly (which would satisfy security but violate the performance and cost constraints), controls are concentrated at zone boundaries and scaled to the sensitivity of what they protect - the customer-data VLAN gets the strongest isolation and monitoring, while general employee traffic passes through segmentation but not redundant deep inspection at every hop.

This selective-strength approach is what allows the design to satisfy security and compliance (non-negotiable) simultaneously with performance and cost (bounded) - segmentation itself is a low-overhead control (VLAN tagging and ACLs add negligible latency) that does most of the containment work, reserving the more expensive deep-inspection resources (IDS/IPS) specifically for the boundaries that matter most.

## **6. Solution Accuracy & Feasibility**

The design is operationally feasible with standard enterprise networking and security equipment (managed switches supporting VLANs/802.1Q, a next-generation firewall with IPS, an IDS sensor, a RADIUS/MFA server, and a SIEM platform such as Splunk or an open-source ELK-based stack) - all mature, widely deployed technologies in financial-sector environments, satisfying real-world compliance frameworks such as PCI-DSS's network-segmentation and access-control requirements. The single internal firewall/IDS pair at the critical boundary, rather than per-VLAN dedicated appliances, keeps capital cost bounded while still meeting the security requirement where it matters most.

## **7. Creativity & Innovation**

The innovative element is the proportional, risk-weighted application of security controls - rather than a flat 'secure everything equally' approach, the architecture concentrates cost and performance overhead exactly where breach impact is highest (the customer-data VLAN), which is both more cost-efficient and, from an audit perspective, easier to justify to regulators as a deliberate, risk-based control design rather than an ad hoc one.